Avikar S. Khakh

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Developer experienced in full-stack development, AI-powered systems, and native mobile, with efficient use of agentic AI to accelerate development. Passionate about translating business needs into working products that deliver real user impact.

EDUCATION

Wake Forest University — May 2026
B.S. Computer Science, Minor in Economics — Winston-Salem, NC

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Software Engineering, Computer Vision
Languages: English (fluent), Spanish (conversational), Punjabi (conversational)

EXPERIENCE

Steward — Winter 2025 – Present
Software Engineer — Winston-Salem, NC

- Sole engineer for a **WFU StartUp Lab-backed** fintech startup (**1 of 9** selected), building a full-stack iOS app that recommends the optimal credit card at every merchant using real-time location
- Developed a recommendation engine scoring cards across **6 reward dimensions** with fuzzy merchant name matching for accurate category detection
- Engineered automated scrapers for **8 major card issuers** (Amex, Chase, Capital One, Citi, and more) that refresh offer data daily, plus a three-layer merchant cache that reduced Google Places API calls by **90%**

NYC Administration for Children's Services — Summer 2025
Software Development Intern — New York, NY

- Designed and deployed an **AI-powered** knowledge search system using **Azure OpenAI** and **RAG** across agency documents for **200+ staff**
- Enhanced **Angular** frontend components and optimized backend API endpoints, improving UI responsiveness by **30%**, reducing page load times by **1.2s**
- Gathered requirements from non-technical stakeholders, translated business needs into code, and shipped enhancements that reduced query response times by **15%**

PROJECTS

Portfolio Strategy Simulator | *Python, FastAPI, React, cvxpy, yfinance* | *GitHub* — Spring 2026

- Built a full-stack portfolio optimization tool (**FastAPI**, **React**, **cvxpy**) with configurable constraints for volatility, turnover, and tracking error
- Engineered a backtesting engine that strictly uses only historically available data at each rebalance point, with **automatic constraint relaxation** to gracefully handle infeasible scenarios
- Created an interactive **React dashboard** with NAV charts, allocation breakdowns, strategy comparisons, and the ability to save and reload past runs

LEADERSHIP

Wake Forest University Robotics Organization — Fall 2024 – Present
President — Winston-Salem, NC

- Scaled the organization from **3 to 20+** active members through hands-on project-based learning and robotics competitions, and grew funding from **$300 to $1,100** by presenting to faculty and the student budgeting office
- Partnered with the CS department to establish a **1.5-credit robotics course** built on the club's curriculum and competition work

TECHNICAL SKILLS

Languages: Python, Swift, JavaScript, TypeScript, Java, C, SQL

Frameworks & Tools: React, SwiftUI, FastAPI, Angular, Tailwind CSS, Docker, PostgreSQL, Git, GitHub, pytest

AI/ML: RAG pipelines, Azure OpenAI, scikit-learn, cvxpy, pandas, agentic workflows (Claude Code, Codex)